

Global Importance Ranking: Extracting Top 10 Titles via TF-IDF

Data Science and Knowledge Discovery

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1 Concept: Unsupervised Global Ranking

When no query is provided, the goal is to find titles that are most descriptive of the specific themes in the database. We rank titles based on their **Global Weight Magnitude**. This identifies titles that contain rare, high-value terms rather than common, low-information words.

2 Mathematical Approach

To rank titles without a query, we calculate a **Title Score** (S_d) for each document d in the corpus.

2.1 Step 1: TF-IDF Matrix Construction

We first compute the weight for every term t in every title d :

$$w_{t,d} = TF(t,d) \times \log\left(\frac{N}{DF_t}\right) \quad (1)$$

2.2 Step 2: Aggregate Title Scoring

We assign a score to each title by aggregating the weights of its constituent terms. Common methods include:

- **Summation:** $S_d = \sum_{t \in d} w_{t,d}$ (Favors longer titles with many unique words).
- **Mean (Centroid):** $S_d = \frac{1}{|d|} \sum_{t \in d} w_{t,d}$ (Normalizes for length, finding the most "dense" titles).

2.3 Step 3: Vector Magnitude (L2 Norm)

Alternatively, the importance can be viewed as the magnitude of the title vector in the feature space:

$$S_d = \|\vec{V}_d\|_2 = \sqrt{\sum_{i=1}^n w_{i,d}^2} \quad (2)$$

3 Database Extraction Workflow

To retrieve the top 10 titles from your database, follow this logic:

1. **Initialization:** Fit a TF-IDF Vectorizer on the entire column of titles in your database.
2. **Weighting:** Transform the titles into a sparse matrix of weights.
3. **Scoring:** For each row in the matrix, calculate the sum or the L2 norm of the weights.
4. **Sorting:** Sort the database records by this calculated score in descending order ($S_d \downarrow$).
5. **Extraction:**

$$\text{Result} = \text{Top}_{10}\{\text{Title}_d \mid \text{argmax}_{S_d}\} \quad (3)$$

4 Comparison of Methods

Metric	Formula	Outcome
Summation	$\sum w_{t,d}$	Highlights long, detailed titles with many specific keywords.
Mean	$\text{avg}(w_{t,d})$	Highlights concise titles that use highly specific terminology.
L2 Norm	$\sqrt{\sum w^2}$	Balanced approach commonly used in Euclidean geometry.

Table 1: Global Ranking Strategies